Problem A: Optimizing Program

The ICPC finals will be held soon, so Yergeldi and his team needs your help. While they were preparing for the competition, they faced an interesting task. You have a list of length N which consists of arrays of given lengths. You have one single operation, you can take any two arrays and merge them into one, the cost of the operation is equal to the sum of their lengths. As a result, you will have a list of N - 1 arrays. The process repeats until there is only one final array left. Find out for what minimum cost it is possible to combine all arrays.

Input format

The first line contains an integer n ($1 \leqslant n \leqslant 2 \cdot 10^5$), the size of the list a. The next line contains n positive integers $a_1, a_2, ..., a_n$ ($1 \leqslant a_i \leqslant 2 \cdot 10^5$), representing the sizes of arrays in the list a.

Output format

Print a single integer - the minimum cost of operations.

Examples

Input

```
4
6 5 3 9
```

Output

```
45
```

Input

```
10
42 18 63 26 19 15 11 29 26 24
```

Output

```
869
```

Notes

Explanation for the first test case:

[6, 5, 3, 9] –> First, merge arrays of lengths 5 and 3 that will cost 8.

[6, 8, 9] –> Next, merge arrys of lengths 6 and 8 that will cost 14.

[14, 9] –> Finally, merge the remaining two arrays that will cost 23.

Therefore, the total cost for merging all arrays is 8 + 14 + 23 = 45.

Problem B: Rock Game

We have a collection of rocks, each rock has a positive integer weight.

With each turn, we choose the two **heaviest** rocks and smash them together. Suppose the stones have weights x and y with $x \leqslant y$. If $x = y$, both stones are totally destroyed. If $x \neq y$, the stone of weight x is totally destroyed, and the stone of weight y has a new weight $y - x$.

At the end, there is at most 1 stone left. Output the weight of this stone (or 0 if there are no stones left).

Input format

The first line contains integer N ($1 \leqslant N \leqslant 10^5$) - the number of stones.

The second line contains N integers a_i ($1 \leqslant a_i \leqslant 10^9$) - the weights of each stone.

Output format

Print the weight of the remaining stone, or 0 if no one left.

Examples

Input

```
6
2 7 4 1 8 1
```

Output

```
1
```

Notes

We combine 7 and 8 to get 1 so the array converts to [2,4,1,1,1] then, we combine 2 and 4 to get 2 so the array converts to [2,1,1,1] then, we combine 2 and 1 to get 1 so the array converts to [1,1,1] then, we combine 1 and 1 to get 0 so the array converts to [1] then that's the value of last stone.

Problem C: Standard problem about soccer

Kakyoin loves football and he goes to the final of the World Cup. At the stadium, he noticed that there are n rows, each can accommodate a distinct number of people. The price of the ticket depends on the row. If there are k ($k > 0$) free seats in the row, then the price of one ticket will be equal to k. What is the maximum amount of money stadium management can get if there are x people in line for a ticket?

Input format

The first line consists of n and x ($1 \leqslant n, x \leqslant 10^5$). n denotes the number of seating rows in the stadium and x denotes the number of football fans waiting in line to get a ticket for the match.

Next line consists of n space separated integers $a_1, a_2, a_3, ..., a_n$ where a_i ($1 \leqslant a_i \leqslant 10^5$) denotes the number of empty seats initially in the i-th row.

It is guaranteed that there are enough free seats for all visitors.

Output format

Print one integer - the maximum amount of money the stadium can earn.

Examples

Input

```
3 10
6 8 9
```

Output

```
67
```

Input

```
1 2
5
```

Output

```
9
```

Notes

The answer may exceed the maximum value of a 32-bit integer — use 64-bit integers.

Problem D: Experiment with Mixtures

Mark is making an experiment with mixtures of different densities. For his experiment he wants all of the mixtures to have a density $\geqslant m$.

He uses the following formula to obtain a combined mixture: $d_{new} = d_{least} + 2 \cdot d_{second\ least}$, where d_{new} is the density of the new mixture, d_{least} is the smallest density among all mixtures and $d_{second\ least}$ is the second smallest density among all mixtures.

Mark repeats the mixing until he gets all the mixtures with the density $\geqslant m$.

You are given the densities of mixtures. How many times Mark should mix his mixtures to get the densities of all mixtures $\geqslant m$?

Input format

The first line consists of integers n and m ($1 \leqslant n \leqslant 10^6, 0 \leqslant m \leqslant 10^9$), the number of mixtures and the minimum required density correspondingly.

The next line contains n space-separated integers d_i ($0 \leqslant d_i \leqslant 10^9$) describing the densities of mixtures.

Output format

Print the number of operations that are needed to make all the densities $\geqslant m$. If it is impossible, print -1.

Examples

Input

```
3 10
1 1 1
```

Output

```
-1
```

Input

```
6 7
1 2 3 9 10 12
```

Output

```
2
```

Notes

The density of a combined mixture can reach about $3 \cdot 10^9$, which does not fit into a 32-bit integer type — use 64-bit integers.

Submit a solution for E-K-th sum

Time limit:	1 s
Real time limit:	5 s
Memory limit:	256M

Problem E: K-th sum

Nurdana loves cookies! She stores all the cookies in separate boxes, but mom allows her to store only k boxes. Her dad is an assistant in this matter. He gives her a box in which there are n cookies. Help her calculate how many cookies she can have?

Input format

The first line contains two integers q and k ($1 \leq q, k \leq 10^5$). Each of the following q lines contains one command.

There are two types of commands:

- *insert* n - dad gives a box with n cookies ($0 \leqslant n \leqslant 10^9$)
- *print* - print the maximum number of cookies Nurdana can have

Output format

For each query of type *print* output, on a separate line, the sum of the k largest of the numbers inserted so far.

If fewer than k numbers have been inserted, output the sum of all of them; if nothing has been inserted yet, output 0.

Equal values are counted separately: if the value 5 was inserted three times and $k \geqslant 3$, all three copies contribute to the sum.

Examples

Input

```
6 4
print
insert 9
insert 6
print
insert 10
print
```

Output

```
0
15
25
```

Examples

Input

```
6 4
print
insert 9
insert 6
print
insert 10
print
```

Output

```
0
15
25
```

Input

```
7 2
insert 2
insert 6
insert 3
print
print
insert 1
print
```

Output

```
9
9
9
```

Notes

This problem must be solved using a heap.

Use long long type for the sum of elements.

Since k is fixed, you don't need to store all numbers, only k largest elements.

The answer can be as large as $10^5 \cdot 10^9 = 10^{14}$, so a 32-bit integer is not enough.